Solving quadratic equations by factoring

1

a $x = 0, x = -6$

b $m = 0, m = -2$

c $x = 0, x = \frac{12}{7}$

d $x = 7, x = 6$

e $x = 7, x = -8$

f $x = \frac{19}{17}, x = \frac{5}{11}$

g $x = -\frac{19}{7}, x = \frac{5}{2}$

h $x = \frac{17}{4}, x = -\frac{11}{3}$

i $x = \frac{9}{4}$

j $x = 1$

k $m = 0, m = 10$

2

a $x = 0, x = -12$

b $y = 0, y = \frac{1}{5}$

c $x = 0, x = -4$

d $x = -10, x = -9$

e $x = -11, x = 5$

f $x = 9, 8$

g $x = 10$

h $x = 7, x = -2$

i $x = 2, x = -4$

j $x = -4, x = 5$

k $x = -\frac{2}{5}, x = -4$

l $x = \frac{9}{4}, x = 4$

m $x = \frac{5}{4}, x = 3$

n $x = -\frac{5}{4}, x = -3$

o $x = -\frac{4}{5}, x = -2$

p $x = -\frac{5}{3}, x = 4$

q $x = -8, x = 8$

r $x = \frac{8}{9}, x = \frac{-8}{9}$

3

a $x = 0, x = 32$

b 32 cm

4

a $m = 14, 13$

b $m = 5, m = -4$

c $m = 0, m = 14$

d $m = 0, m = \frac{7}{11}$

e $y = -1, y = 3$

f $x = \frac{-5}{2}, x = 12$

5

a -10

b 1

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7 $x = 2, x = -3$

8

a $t = 4, t = 8$ **b** $4, 4$

9 $k = -16$

10 $a = 13$

11

a No **b** No **c** No **d** Yes

Additional questions

12

a

i $(0,\, 6)$ **ii** $(-1,\, 0)\,,\, (6,\, 0)$

iii $y = -(x - 6)(x + 1)$

b

i $(0,\, 12)$ **ii** $(2,\, 0)\,,\, (6,\, 0)$

iii $y = (x - 2)(x - 6)$

c

i $(0,\, -16)$ **ii** $(4,\, 0)$

iii $y = -(x - 4)^2$

d

i $(0,\, 0)$ **ii** $(-5,\, 0)\,,\, (0,\, 0)$ **iii** $y = x(x + 5)$

13 10 yards

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